

IBE; discussion

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→ How to impersonate?

- ⇒ Same ID is not sufficient
- ⇒ Must «register» to PKG to get private key

→ PKG: very powerful entity

- ⇒ Solution: Distributed PKG!
 - Use pedersen's DKG
- ⇒ Intermediate approaches: certificateless crypto
 - See next slide

→ Multi-authority?

- ⇒ Yes, several extensions

ECQV

→ **Elliptic Curve Qu-Vanstone**

- ⇒ A.k.a. implicit certificates
- ⇒ Identity-based

→ **Idea: self-certifying EC point**

- ⇒ Do not use PKG but use a CA
- ⇒ But CA does **not** issue standard certificates
- ⇒ it «helps» to extract the user public Key from an «implicit» identity-based certificate

→ **Advantage: very small (implicit) certificates!!**

- ⇒ A single EC group point!

Initial registration

→ **CA:**

⇒ Private key → c

⇒ Public key → $C = cG$

Before asking for a certificate, the device I_A generates a random positive integer r_A ; then, it computes a point R_A on the chosen elliptic curve, $R_A = r_A \cdot G$, and sends it to the CA. The CA extracts a random positive integer k , computes $k \cdot G$ (another point on the elliptic curve), generates the implicit certificate, P_A , and the implicit signature, γ_A , by using:

$$\begin{cases} P_A = R_A + k \cdot G \\ \gamma_A = c + k \cdot H(P_A, I_A) \end{cases} \quad (1)$$

Now, it sends them to the device. Then, it can generate its private key, $s_{v,A}$, and its public key, $P_{b,A}$, by:

$$\begin{cases} s_{v,A} = \gamma_A + r_A \cdot H(P_A, I_A) \\ P_{b,A} = s_{v,A} \cdot G \end{cases} \quad (2)$$

verification

As well-known, the most important strength of ECQV is that the public key of a given device can be computed by any other third-party starting from the knowledge of the implicit certificate and the public key of the authority:

$$P_{b,A} = s_{v,A} \cdot G = C + P_A \cdot H(P_A, I_A). \quad (3)$$

→ **In conclusion:**

⇒ Unlike IBE PKG, no third party knows user private keys

→ **Certificates: reduced to single (!) group point!!**

⇒ Very low transport overhead compared to traditional certificates

→ $O(40)$ bytes versus $o(1000+)$ bytes

→ **Application in low-resources IoT/wireless networks**

⇒ See e.g., Sciancalepore, Capossele, Piro, Boggia, Bianchi, *Key management protocol with implicit certificates for IoT systems*. 2015 Workshop on IoT challenges in Mobile and Industrial Systems (pp. 37-42).

⇒ Used (at that time) only in ZigBee: **Certificate Based Key Establishment (CBKE) protocol**